

Progressive gate-unfreezing of the QFT operator

One gate thawed per stage, trained to a plateau — three unfreeze orderings $\times$ two initialisations on DIV2K-8q

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Question

The **QFTBasis(8, 8)** (72 $U(2)$ gates) starts either at the QFT-family **identity** ($T_{t=0}(x) = x$) or **Haar-random**. We **unfreeze one gate per stage**, train it to a plateau, then thaw the next. Does the **order** — block-growth (bg), left $\rightarrow$ right (lr), right $\rightarrow$ left (rl) — or the **initialisation** change the trajectory or the endpoint?

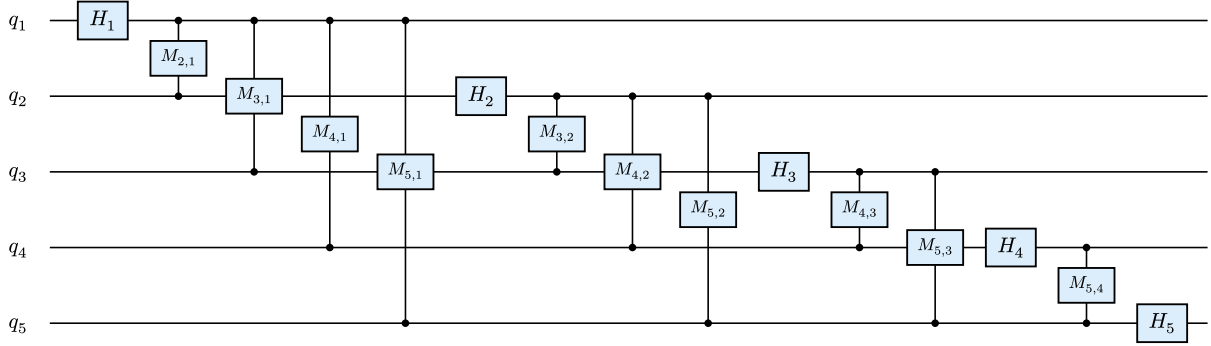


Figure 1: The QFT(5) circuit (one axis) with every gate named: H_i = Hadamard on q_i ; $M_{j,i}$ = controlled-phase between q_j and q_i ($j > i$). The three orderings thaw these same gates in the sequences below; the $m = 8$ runs use the analogous 72-gate circuit.

The three unfreeze orderings traverse the gates of Figure 1 as:

- **block-growth** (bg): finish QFT(k) before adding qubit $k\{+\}1$ — H_1 ; $H_2, M_{2,1}$; $H_3, M_{3,1}, M_{3,2}$; $H_4, M_{4,1}, M_{4,2}, M_{4,3}$; $H_5, M_{5,1}, \dots, M_{5,4}$. Long-range couplings come last.
- **left $\rightarrow$ right** (lr): QFT construction order — $H_1, M_{2,1}, M_{3,1}, M_{4,1}, M_{5,1}, H_2, M_{3,2}, \dots, H_5$.
- **right $\rightarrow$ left** (rl): the reverse — H_5 first, H_1 last.

Setup

Fixed batch of 50 training images; **MSELoss** on the top-10% coefficients. pdft's JIT Adam step runs with a per-stage frozen set (moments reset per stage); a Riemannian grad-norm probe gives the plateau signal. A stage ends at $\|g\| < 10^{-5}$ or $|\Delta L| < 10^{-5}$ (min 5 steps), else an 800-step cap. Random init: Haar $U(2)$ per Hadamard, uniform phase per controlled-phase, seed shared across orderings.

Training dynamics

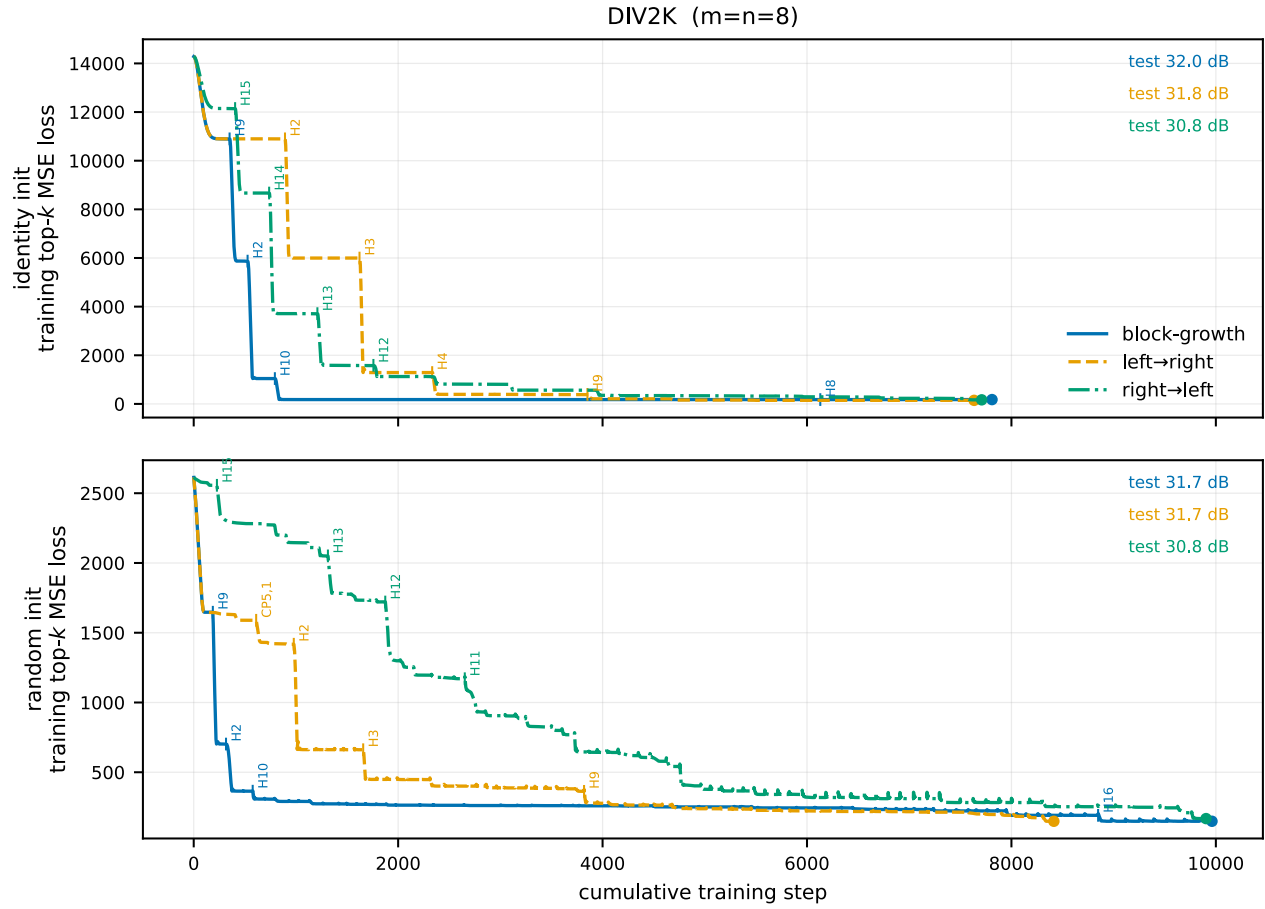


Figure 2: Absolute **training** top- k MSE loss vs cumulative step, one curve per ordering (rows: identity / random init). Tick labels mark **which gate** (e.g. H7, CP3,1) was thawed at each of the largest drops. The y-axis is the **training** top-10%-coefficient MSE loss; endpoints are annotated with the separate **test PSNR@ $\rho\{=\}.20$** (image reconstruction on held-out data) — the two are different metrics, so the curve’s height does not determine the PSNR. Per-cell loss+grad-norm staircases: [div2k_8q/<init>/figures/](#).

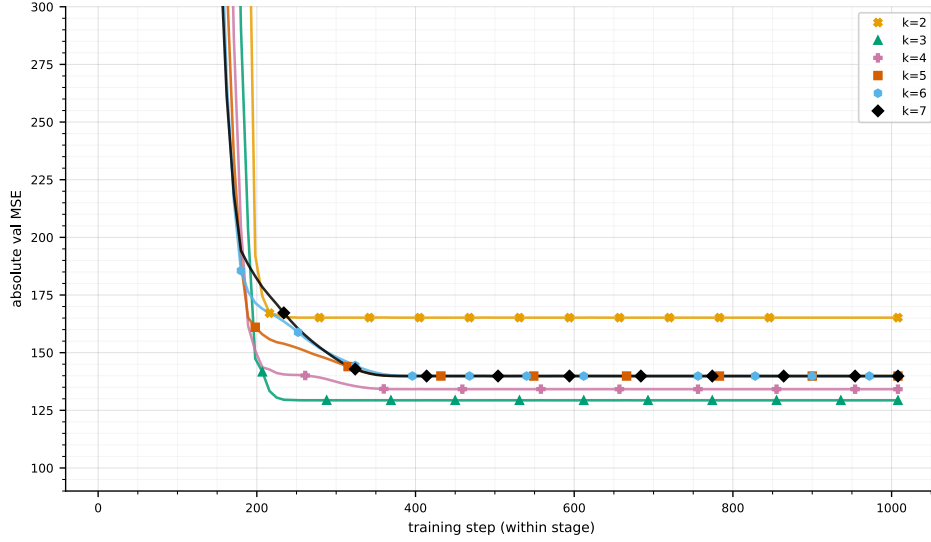


Figure 3: **Block-size** curriculum (`qft_progressive`, DIV2K) for comparison: per-stage validation MSE for block sizes $k = 2..7$, each one whole $2^k \times 2^k$ QFT trained 1008 steps. Eight whole-block stages here vs the 72 one-gate-at-a-time stages above — same ≈ 31.7 dB endpoint (table below).

Reconstruction PSNR (DIV2K)

Gate-unfreeze endpoints **together with** the `qft_progressive` block-size baseline (train all $k(k+1)$ gates of a $2^k \times 2^k$ QFT at once, per stage k). DIV2K test PSNR (dB) at four keep ratios ρ :

configuration	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$
unfreeze · identity · bg	19.4	26.9	29.7	32
unfreeze · identity · lr	25.2	27.9	30	31.8
unfreeze · identity · rl	24.8	27.2	29.1	30.8
unfreeze · random · bg	25.2	27.8	29.8	31.7
unfreeze · random · lr	25.2	27.8	29.8	31.7
unfreeze · random · rl	24.8	27.2	29.1	30.8
block-size · k=1 (2×2)	9.4	12.5	16.2	21.1
block-size · k=2 (4×4)	19.1	26.9	29.8	32
block-size · k=3 (8×8)	25.2	28.1	30.3	32.3
block-size · k=4 (16×16)	25.3	28	30.1	32
block-size · k=5 (32×32)	25.2	27.8	29.8	31.7
block-size · k=6 (64×64)	25.2	27.8	29.8	31.7
block-size · k=7 (128×128)	25.2	27.8	29.8	31.7
block-size · k=8 (256×256)	25.2	27.8	29.8	31.7
classical · block-DCT 8×8	26.1	29.4	31.9	34
classical · block-FFT 8×8	24.5	27.1	29.1	30.8

All learned QFT schemes reach the **same** QFT(8,8) endpoint, ≈ 31.7 dB @ $\rho\{=\}.20$: the block-size curriculum saturates by $k=2$ and the gate-unfreeze sweep lands there from either init and any order — **schedule-independent** (rl trails ≈ 1 dB; every gate-unfreeze stage ended on the loss- Δ plateau, none at the step cap). For reference, the best **any** prior trained basis reaches is 33.7 dB (rich/identity). The classical **block-DCT 8×8** is in fact the strongest here (34 dB @ $\rho\{=\}.20$), ahead of every learned QFT basis; **block-FFT 8×8** is weaker (30.8 dB) — the QFT-family operators sit between the two classical block transforms.

Reading

The staircase shows each gate’s marginal value: the big drops are the **early** gates (the marked Hadamards, e.g. H9, H2 for block-growth); later gates only fine-tune. One caveat on the endpoint labels — the train-loss order

need not match the test-PSNR order (**bg** has the **highest** train MSE yet the **best** test PSNR): the loss is a top-10%-coefficient proxy on the train batch, PSNR a 20%-keep reconstruction on the test set.